Fiscal Year:	FY 2005	**Task Last Updated:**	FY 07/21/2005
PI Name:	Maida, James C M.S.		
Project Title:	Enhanced Lighting Techniques and Augmented to Improve Human Task Performance		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Space human factors engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	01-OBPR-07
Start Date:	11/01/2002	**End Date:**	10/31/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	Shuttle/ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	“On orbit” tasks involve procedures that are very complex and highly dependent on the availability of visual information. In many situations, cameras are used as tools to help overcome the visual and physical restrictions associated with space flight. However, these cameras are affected by the dynamic lighting conditions of space, particularly during critical robotic and EMU operations. Normal training for these tasks includes computer generated simulations of planned camera views. Improvement of current trainer design using enhanced visual display information (dynamic overlays or augmented reality) and the use of shadowing in computer-aided training (a major enhancement to current simulators) are candidates for improving human performance. Because of the potentially significant costs of implementation in production trainers, it is important to accurately determine the degree of improvement. In the case of dynamic overlays or augmented reality, the proposed project will perform subject testing in the Multiuse Remote Manipulator Development Facility at Johnson Space Center (MRMDF). The project will investigate the use of		

	augmented reality to assist payload berthing where alignment guides are structural elements not necessarily dedicated or known in advance as alignment cues. Current efforts in the MRMDF are focusing on improving grappling operations using augmented reality technology. Using the AR Toolkit from the University of Washington, graphical images and alignment guides are overlaid on live video. The system has been ported to a laptop and uses a USB port for video input. Initial testing is being done in the Robotics Evaluation Laboratory in MRMDF. Additional testing will be performed for the Dexterous Manipulator Trainer and the Space Station Remote Manipulator System Trainer.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Augment reality technology has useful applications where "just in time" information and guidance is important. Firefighters can use this type information in a helmet mounted display while exploring a smokey building interior. This technology is already being used in sports entertainment areas such as the virtual first down line in football.
Task Progress:	Project Status and Preliminary Results. The project has been investigating, in detail, the role of augmented reality in operator performance with the DMT and with the implementation of a reliable and flexible system for testing the effects of AR on performance. A portable laptop based system using the AR Toolkit from University of Washington has been successfully developed and tested for use in experimentation with he DMT. Using pilot testing a system of overlays symbols has been developed and refined for use in experiments with experienced robotics operators at JSC. The overlay symbol system is designed to provide the following advantages: 1) alignment of ORU and receptacle within insertion tolerances using a single camera view, even under adverse lighting conditions, 2) “Rubber band” features prompt appropriate control command motions, even when the destination receptacle is outside the= field of view of the camera and 3) allow separation of needed translation and rotation control inputs. Experimental Protocol. Subject testing protocol features: • Within subjects experimental design comparing performance with and without AR symbols. • The experimental task includes only ORU positioning for insertion in a receptacle. Because of subject testing time and metric constraints grappling and insertion tasks were not included. • A trial begins with ORU in a predetermined position and terminates at the point where the ORU is poised for insertion into the receptacle, and the Force Moment Analysis System is activated. • Four starting positions and fixed task illumination provide both favorable and unfavorable task lighting conditions. • Insertion of the ORU is attempted with +X only hand controller input to provide feedback to the subject as to the level of success of the pre-insertion alignment. Future and On-going Activities. Pilot testing is underway to evaluate experimental protocol, data acquisition and analysis processes and performance metrics for selection of subjects for testing. Subject testing will be conducted through the Spring of 2005. These activities are being coordinated with an NRA funded project entitled "Real-time Augmented Reality (AR) for Space Station Remote Manipulators" which is just underway. If subject testing results are positive, it is anticipated that other on-orbit tasks will be examined for application of this technology for improved human performance.
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